DATA PROFILER

Data Preprocessing and Feature Engineering

Theory Documentation

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This document presents the fundamental concepts of data preprocessing and feature engineering, which are essential steps in any data science project. It covers key topics such as data cleaning, exploratory data analysis, and data profiling in a simple and clear manner.

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1. Introduction to Data Analysis

Data analysis is the process of collecting, cleaning, and studying data to find useful information.

It helps us understand patterns and make better decisions. In real life, companies use data analysis to understand customer behavior and improve their services.

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2. Steps in a Data Science Project

A data science project usually follows these steps:

- Data Collection – Gathering data from different sources
- Data Cleaning – Fixing errors like missing values or duplicates
- EDA (Exploratory Data Analysis) – Understanding data using graphs
- Modeling – Building machine learning models
- Evaluation – Checking model performance

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3. Machine Learning Problem

A machine learning problem means using data to make predictions.

There are two main types:

- Classification – Predict categories
  Example: Will a customer leave (churn: Yes/No)

- Regression – Predict numbers
  Example: Predict income or sales

In this project, predicting customer churn is a classification problem.

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4. Data Collection Methods

Data can be collected from different sources:

- CSV files – Simple table format
- JSON files – Structured data format
- APIs – Data collected from online services

Using multiple sources helps in getting more complete data.

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5. Data Cleaning

Data cleaning is the process of fixing problems in the dataset.

Common issues:

- Missing values – Some data is empty
- Duplicates – Same rows repeated
- Outliers – Extremely high or low values

Cleaning makes the data more accurate and reliable.

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6. Exploratory Data Analysis (EDA)

EDA is used to understand data using graphs and charts.

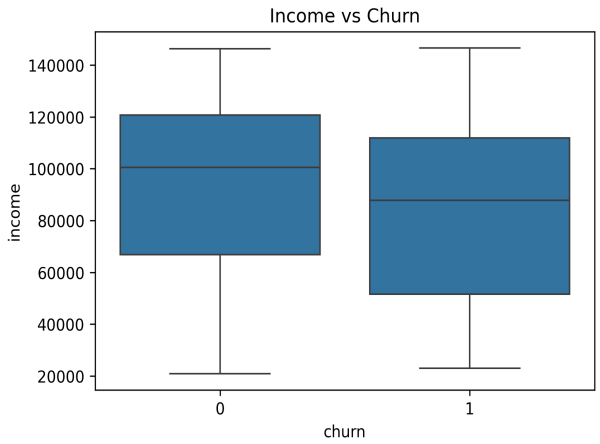
Types of EDA:

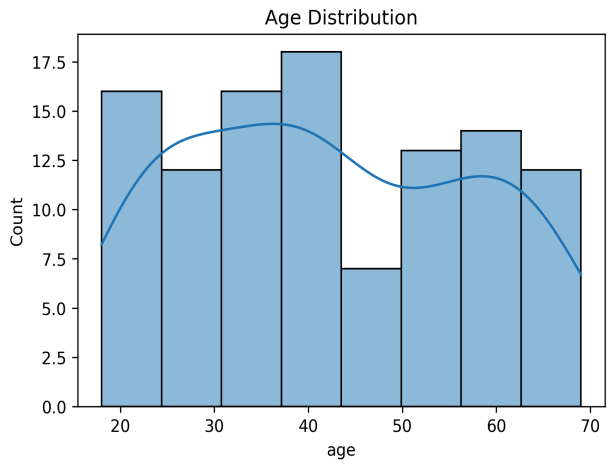
- Univariate Analysis – Study one column
  Example: Age distribution

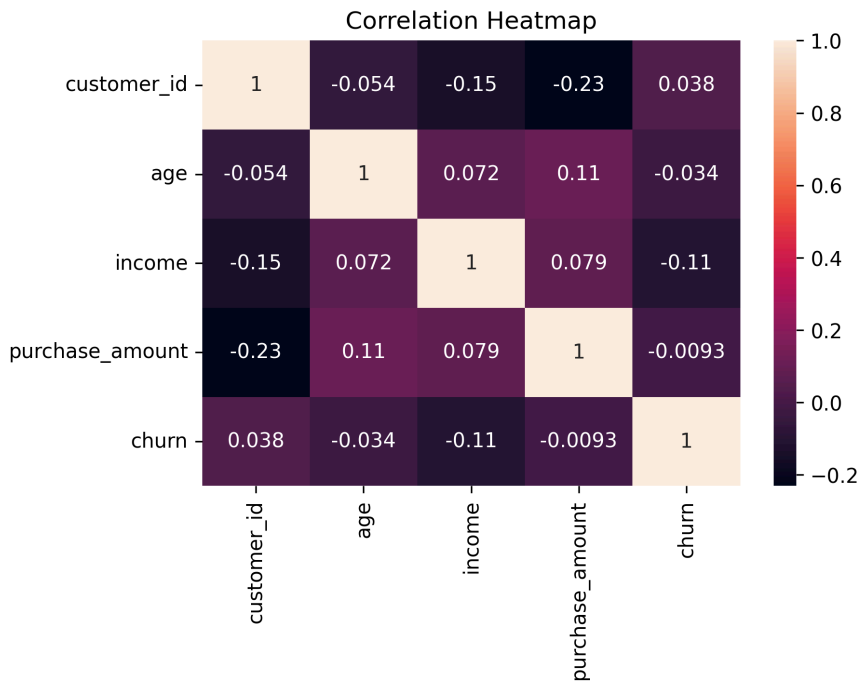
- Bivariate Analysis – Compare two columns
  Example: Income vs Churn

- Multivariate Analysis – Study multiple variables
  Example: Correlation heatmap

EDA helps in finding patterns and relationships in data.







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7. Data Profiling

Data profiling is a process of creating a summary report of the dataset.

It shows:

- Missing values
- Statistics
- Correlations

In this project, we used ydata-profiling to generate an automatic report.

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8. Feature Engineering

Feature engineering means creating or improving features (columns) to make data more useful.

Examples:

- Converting categories into numbers (encoding)
- Scaling values
- Creating new features

This step improves machine learning performance.

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9. Real-World Importance

In real companies, most of the work is not building models but preparing the data.

If the data is not clean:

- Models give wrong results
- Predictions become inaccurate

For example, if customer income is missing or incorrect, the churn prediction model will fail.

That is why data preprocessing and feature engineering are very important in real-world projects.

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10. Conclusion

Data preprocessing and analysis are important steps in any data science project.

By cleaning and understanding data, we can make better predictions and build reliable machine learning models.